

KEYA HU

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<https://lillian039.github.io/>

EDUCATION

Massachusetts Institute of Technology (MIT)

2025.9 – Present

Ph.D. in Computer Science

Advisor: Prof. Kaiming He & Prof. Jacob Andreas

Shanghai Jiao Tong University (SJTU)

2021.9 – 2025.6

B.Eng in Computer Science, Zhiyuan College, ACM Honors Class

PREPRINT

VISTA: A Visual Harness for Reasoning in an Interactive World

Qiushi Han*, Keya Hu*, Linlu Qiu*, Cathy Wu, Kaiming He

in submission

Keywords: Visual harness, Interactive games

PUBLICATIONS

ELF: Embedded Language Flows

Keya Hu*, Linlu Qiu*, Yiyang Lu, Hanhong Zhao, Tianhong Li, Yoon Kim, Jacob Andreas, Kaiming He

NeurIPS, 2026

Keywords: Diffusion Language Models, Flow Matching, Continuous Time and Embedded Space

ARC Is a Vision Problem!

Keya Hu, Ali Cy, Linlu Qiu, Xiaoman Delores Ding, Runqian Wang, Yeyin Eva Zhu, Jacob Andreas, Kaiming He

CVPR, 2026

Keywords: Visual Reasoning, Abstract Reasoning

Combining Induction and Transduction for Abstract Reasoning

Wen-Ding Li*, Keya Hu*, Carter Larsen, Yuqing Wu, Simon Alford, Caleb Woo, Spencer M. Dunn, Hao Tang, Wei-Long Zheng, Yewen Pu, Kevin Ellis

ICLR, 2025, Best paper in ARC-prize 2024

Keywords: Abstract Reasoning, Large Language Model, Data Generation

Code Repair with LLMs gives an Exploration-Exploitation Tradeoff

Hao Tang, Keya Hu, Jin Peng Zhou, Sicheng Zhong, Wei-Long Zheng, Xujie Si, Kevin Ellis

NeurIPS, 2024

Keywords: Program Synthesis, Code Repair, Explore-exploit Tradeoff, Large Language Model

Contrastive Self-Supervised EEG Representation Learning for Emotion Classification

Keya Hu, Ren-Jie Dai, Wen-Tao Chen, Hao-Long Yin, Bao-Liang Lu, and Wei-Long Zheng

EMBC, 2024, oral

Keywords: Self-supervised Learning, Electroencephalography (EEG), Affective Computing

* *Equal contribution*

RESEARCH EXPERIENCE

Research Intern	<i>2026.5 – 2026.9</i>
PHI Agent Team, Apple, CA, United States	
Advisor: Philipp Krähenbühl	
• Unified continuous diffusion model, pretraining	
Visiting Research Intern	<i>2024.1 – 2025.1</i>
Recursion and Learning Lab, Cornell University, NY, United States	
Advisor: Kevin Ellis	
• Efficient Program Synthesis with LLM, Human-Level Abstraction Reasoning	
Research Assistant	<i>2023.9 – 2024.7</i>
Brain-like Computing and Machine Intelligence Lab, SJTU, Shanghai, China	
Advisor: Wei-Long Zheng and Bao-Liang Lu	
• EEG-signals for Emotion Recognition, Self-Supervised Learning	

TEACHING EXPERIENCE

Teaching Assistant	Data Structures	<i>2023 Spring</i>
Teaching Assistant	Programming Design	<i>2022 Fall</i>

AWARDS AND SCHOLARSHIPS

Siebel Scholars	2027
The Siebel Scholars program represents the brightest minds from around the globe.	
Jane Street Graduate Research Fellowship, Honorable Mention	2026
Honorable mention for the Jane Street Graduate Research Fellowship.	
Schwarzman College of Computing Fellowship	2025
One-year fellowship to newly admitted MIT PhD students with exemplary academic achievements.	
Outstanding Graduate of Shanghai	2025
Award for top undergraduate graduates in Shanghai.	
Best Paper Award in ARC-Prize 2024	
Award for best paper in Abstract Reasoning and Corpus(ARC) Kaggle competition	
National Scholarship	2024
Award for 0.2% of students, the highest national-level scholarship.	
Zhiyuan Honors Scholarship	2021, 2022, 2023, 2024
Award for top students majoring in science	
Longhu Scholarship	2023
Award for top students majoring in Computer science	

SERVICE

Reviewer , NeurIPS 2025, 2026
Reviewer , ICML 2026 (<i>Gold Reviewer Award</i>)